

Problem 285: Sky Duel

Difficulty: Easy

Author: UK In-House Applications Team (Abby Watton, Ryan Gaudion, John Vinnell, Simon Wilson, Harvey Whyte, Lucas Osborne, & Dom Kerrigan)

Originally Published: Code Quest 2026

Problem Background

Lockheed Martin is testing two advanced aircraft; the F-22 Raptor, a fast and agile stealth fighter, and the Sikorsky S-97 Raider, a high-speed helicopter. Each aircraft can be assigned to multiple missions; each of these missions has a specified distance to cover and a payload that needs to be carried. Engineers need to determine which aircraft would be most effective for each mission based on their speed, fuel efficiency, payload capacity, and maintenance costs.

Problem Description

Given parameters for each mission, your team will need to determine the best aircraft to use. For each mission, an aircraft can attempt to complete it only if the required payload does not exceed the aircraft's maximum payload limit. Assuming the payload fits within both aircraft's limits, you'll need to use this formula to calculate which is more practical:

$$time = \frac{distance}{speed} * 60 * 60$$

$$fuel = distance \times fuel_rate$$

$$score = time + fuel + maintenance_costs$$

The aircraft with the lower score is considered to be most effective for the mission. In the event the aircraft have very similar scores (a difference of less than 0.01), favor the one that would consume less fuel during the mission. Naturally, if one aircraft cannot carry the mission's payload, the other aircraft should be used. At least one of the aircraft is guaranteed to be viable for each mission.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include:

- A line containing the following positive integers, separated by a space and representing:
 - **N**, the number of missions in the test case
 - The speed (in km/hr) of the F-22 Raptor
 - The fuel rate (in L/km) of the F-22 Raptor
 - The maximum payload weight (in kg) of the F-22 Raptor

- The inspection and maintenance costs for the F-22 Raptor
- The speed (in km/hr) of the Sikorsky Raider
- The fuel rate (in L/km) of the Sikorsky Raider
- The maximum payload weight (in kg) of the Sikorsky Raider
- The inspection and maintenance costs for the Sikorsky Raider
- N lines, each containing information about a mission. Each line contains two positive integers, separated by a space:
 - The distance to be covered during the mission, in km
 - The weight of the required payload, in kg

```
2
2 1200 6 10000 1000 400 1 2000 250
600 1500
3000 1800
3 900 5 5000 500 450 2 2500 100
200 500
1200 1000
300 3000
```

Sample Output

For each test case, your program must identify the most effective aircraft to use for each mission based on the provided information and the calculations given above. For each mission, print a single line containing the name of that aircraft; either “F-22 Raptor” or “Sikorsky Raider”. Address each mission in the order provided.

```
Sikorsky Raider
F-22 Raptor
Sikorsky Raider
F-22 Raptor
F-22 Raptor
```